

Project 2: Formal Criteria-Based Test Design & Automation

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In this project, we are testing a system called the **Premium Discount Eligibility System**.

The system calculates the discount percentage for a customer depending on:

- the customer type
- the number of orders in the last year
- whether the customer is subscribed to the newsletter or not

The class that we will create is: **DiscountCalculator**

The method that will calculate the discount is:

```
calculateDiscount(String customerType, int totalOrdersInLastYear, boolean  
isSubscribedToNewsletter)
```

In this project, we followed the **Model-Driven Test Design** method.

This means that we designed the test cases first before writing the code. We first looked at the inputs of the system, divided them into blocks, checked if there are any impossible combinations, and then created Pair-Wise test cases.

This helps us test the program in an organized way instead of choosing random test cases.

Input Space Partitioning (ISP)

In this project, we applied Input Space Partitioning (ISP) as part of the MDTD methodology. First, we identified the main characteristics that affect the behavior of the discount calculation system. Then, each characteristic was divided into blocks that represent possible groups of input values.

Characteristics and Blocks

Characteristic C1: Customer Type

- Block B1: NEW
- Block B2: REGULAR
- Block B3: PREMIUM

Characteristic C2: Total Orders in Last Year

- Block B1: Less than 10 orders
- Block B2: 10 orders or more

Characteristic C3: Newsletter Subscription

- Block B1: Subscribed
- Block B2: Not Subscribed

The purpose of dividing the inputs into blocks is to reduce the number of needed test cases while still covering different possible behaviors of the system.

Pair-Wise Coverage (PWC)

After identifying the characteristics and their corresponding blocks, Pair-Wise Coverage (PWC) was applied to generate test combinations. The goal of PWC is to ensure that every possible pair of blocks between two characteristics appears in at least one test case.

Pair-Wise Combinations

Test Case	C1: Customer Type	C2: Total Orders	C3: Newsletter Subscription
TC1	B1 (NEW)	B1 (<10)	B1 (Subscribed)
TC2	B1 (NEW)	B1 (<10)	B2 (Not Subscribed)
TC3	B2 (REGULAR)	B1 (<10)	B1 (Subscribed)
TC4	B2 (REGULAR)	B2 (≥10)	B2 (Not Subscribed)
TC5	B3 (PREMIUM)	B1 (<10)	B2 (Not Subscribed)
TC6	B3 (PREMIUM)	B2 (≥10)	B1 (Subscribed)

Infeasible Combination

Test Case	C1: Customer Type	C2: Total Orders	Reason
TC7	B1 (NEW)	B2 (≥10)	A NEW customer cannot have 10 or more orders in the last year

This infeasible combination should result in an `IllegalArgumentException` according to the system requirements.